# Can Gliner2 directly identify entities and their political affiliation ?

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## Abstract

We apply the GLiNER2 model to identify a candidate and his political affiliation based on a transcript of his manifesto in the 1988 parliamentary election in France. We explore several different specifications of this task and compare the recall performance. In particular, we are interested in the structured extraction feature of GLiNER2, in comparison with standard NER or classification tasks. The performance of featured extraction is quite high and only slightly lower than standard NER, suggesting that structured extraction will be useful for future projects at INSEE, such as for mapping free-text passages into official categories.

## 1 Introduction

Recent advances in natural language processing has made extraction of structured information from textual data more flexible and user friendly. Current Large Language Models (LLMs) being pre-trained and general purpose, complex tasks can be queried with a simple prompt without the need for computer-intensive model-training or labour-intensive annotation of training sets.

As a civil servant at INSEE, I am very interested in the structured extraction features of GLiNER2. This would make it very easy to specify tasks related to nomenclature, e.g. extraction of questionnaire answers from free text, primary sector of companies from their website description, etc. For application at INSEE, we require a model that runs quickly on massive datasets, with limited supervision and a user-friendly specification of the task.

### 1.1 State of the art

Current large language models (Claude, ChatGPT, Gemini, etc.) can achieve very high performance in various text processing tasks. However, as highlighted in the course lecture notes, they have several drawbacks that limit their usability for INSEE's purposes. For instance, the model may provide an answer that does not follow the intended schema. As we would like to run automated queries on large samples, this is not acceptable. Additionally, we would like guarantees on the fact that the model reads the complete input. We tested the recently released `gpt-oss:20B` [1] and `gemma4:26B` [2] models, but we were unable to make them work smoothly on our Linux laptop with 32GB RAM (no GPU).

The course also introduced spaCy [3] which is a lightweight convolutional neural network dedicated to specific tasks such as named entity recognition. This model is very fast, but does not allow structured extraction. Hierarchical entity extraction would not be possible.

GLiNER2 [4] appears to provide the best of both worlds. It is a transformer-based model optimized for a few tasks including NER and classification, and we can very easily specify the type and structure of the information to be extracted. However, it is not clear if the provided models have been trained for French documents. Also, there are several possible ways to define the task and we need benchmarks to select the best performing specification.

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### 1.2 Objective of the project

In this project, we want to evaluate the feasibility and performance of using a LLM for a user-defined Named Entity Recognition task. We use Gliner2 to extract entities, i.e. names, from the manifestoes of candidates in the 1988 parliamentary elections in France available in the Archelec corpus.

In particular, we want to evaluate the LLM performance in extracting the political affiliation of the main candidate for several different specifications of this task. We use the provided metadata to partially annotate the manifesto. We then compute the recall measure when the task is specified as a classification task, a NER task with one entity per label, or as a structured extraction task with political affiliation as a required field.

## 2 The Archelec dataset

The Archelec corpus [5] is a collection of documents consisting of the manifestoes of candidates in elections in the French Fifth republic. The documents were collected and digitised by Sciences Po.

In this project, we focus on the 1988 parliamentary election, for which the transcripts and metadata were made available in this course, around 3,400 documents in total. We include both first and second-round manifestoes because we do not expect significant biases linked to first or second round in this project. For instance, we expect all the major political parties to remain represented among the second round candidates.

The transcripts are provided as plain text files. They were obtained by OCR, and therefore can contain spurious elements (typically "Sciences Po" or other headings, footers, etc.) as well as transcription errors. Metadata is provided as a CSV file. We will mostly use the `titulaire-nom` and `titulaire-soutien` columns.

### 2.1 Political affiliation

Political affiliation of the candidate is not straightforwardly provided in the metadata. Instead, the `titulaire-soutien` column contains a list of supporting political parties for each candidate. We eliminate candidates without support, or who are supported exclusively by non-aligned or "Indépendants" parties. However, many candidates have several supports and we need to summarize their affiliation.

We compute the political affiliation or position of a manifesto as a numerical summary of the list of supports. First, we classify political parties into broad categories according to Wikipedia[1] and arbitrarily attribute a number between 0 (left) and 1 (right) to the category. We then compute the average of the positions for each manifesto. Table 1 shows the main patterns for the political support of a candidate. Most candidates are supported by a single party, but some have multiple supports and are positionned at the average position. Note that unaffiliated parties are removed.

## 3 Visualization of documents and true political affiliation

In this section, we represent the transcripts in a low-dimensional space in order to evaluate the difficulty of political party prediction as a classification task. If the parties are well separated, we expect that GLiNER2 will easily infer political affiliation and that poor performance is related to name recognition rather than affiliation recognition.

We use a straightforward approach. We first train a Doc2Vec [6] model on the transcripts, then apply t-SNE [7] on the model embeddings. Figure 1 shows the transcripts in the t-SNE space, with colors corresponding to the political position computed above. Transcripts with similar position index are clustered, which indicates that the simple Doc2Vec approach can easily capture political affiliation.

[1] `https://fr.wikipedia.org/wiki/Liste_des_partis_politiques_en_France_sous_la_Cinqui%C3%A8me_R%C3%A9publique`

Table 1: Main political support patterns and corresponding political position. Patterns with less than 10 transcripts are not shown, and unaffiliated supports are filtered out. Most candidates have a single support, but multiple supports can lead to an intermediate position on the political spectrum.

| List of supports | Political position | Number of candidates |
|---|---|---|
| Extrême gauche | 0.00 | 14 |
| Gauche | 0.20 | 747 |
| Centre gauche, Gauche | 0.30 | 13 |
| Verts | 0.40 | 11 |
| Centre droit, Centre gauche, Gauche | 0.40 | 26 |
| Centre droit | 0.60 | 37 |
| Centre droit, Centre droit | 0.60 | 13 |
| Centre droit, Droite | 0.70 | 374 |
| Centre droit, Droite, Droite | 0.73 | 106 |
| Droite | 0.80 | 40 |
| Extrême droite | 1.00 | 538 |

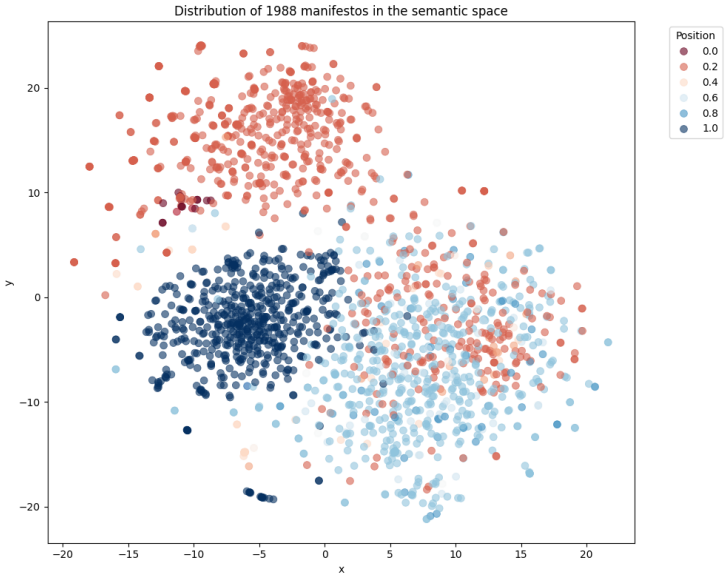


Figure 1: Visualization of the manifestos in the embedding space using Doc2Vec and t-SNE. Each point represents one transcript and the color indicates the political position. Transcripts belonging to similar political parties are close, which means that the transcript contains information on the political affiliation and that Doc2Vec captures it.

# 4 Methods

## 4.1 GLiNER2 tasks

We define several tasks for GLiNER2 in order to evaluate its capabilities.

First, we establish a baseline using the standard Named Entity Recognition categories PERSON, ORGANIZATION and LOCATION. This allows us to assess the performance of GLiNER2 in distinguishing the candidate names from other types of entities (PERSON as opposed to LOCATION).

We also define affiliated entities `PER_LEFT` and `PER_RIGHT` to ask GLiNER2 to recognize political affiliation. As the GLiNER2 documentation does not specify which languages were used during training, we also define the entities `PER_GAUCHE` and `PER_DROITE` with the same meaning to check if the specification language affects the results.

Third, we define a classification task to assess if GLiNER2 can recognize the overall affiliation even if it is not able to attribute it to a person.

Finally, we ask GLiNER2 to extract the information as a structure, i.e. we ask for all persons in the transcript with their name and political affiliation.

The final specification is indicated below. Note that classification and structured extraction are document-level tasks, whereas entity recognition are at entity-level.

```
schema = (extractor.create_schema()
    .entities({
        "PER": "person",
        "PER_LEFT": "person affiliated with left wing political parties",
        "PER_RIGHT": "person affiliated with right wing political parties",
        "PER_GAUCHE": "personne affiliée avec un parti politique de gauche",
        "PER_DROITE": "personne affiliée avec un parti politique de droite",
        "ORG": "organization",
        "LOC": "location"
    })
    .classification("political ideology", ["left","right"])
    .structure("person")
        .field("name", dtype="str", description="name of the person")
        .field("political affiliation", dtype="str", choices=["left","right"])
)
```

### 4.2 Constructing an evaluation set for named-entity-recognition

The provided metadata is at the document level. To evaluate the performance on NER tasks, we need to build entity-level annotations, i.e. to label spans of text with the corresponding name.

We expect the name of the main candidate and his running mate to appear in the transcript, in the main text or as contextual elements. Both should be retrieved by the OCR. We expect that the name of the candidate appears more often in the main text than the name of the running mate, and therefore focus on recognition of the main candidate. We also expect that the name is present and recognized by the NER method in various forms, with or without the firstname for example, so we focus on the last name only.

Consequently, we detect all the appearances of the last name of the main candidate in the transcript. Names are normalized by converting to lowercase and converting accents to their nearest ASCII character. A detected NER is considered successful if it contains the normalized name, i.e. if the textual span encompasses the span of the normalized name. Because we detect all the positions where the name appears, we measure the performance of entity recognition independently of OCR performance.

The proposed approach is adequate for measuring recall, i.e. the probability to detect a name among all true names. The model is penalized for false negatives, so we need to construct a list of entities that should not be missed, which is easy based on the name. On the other hand, precision involves evaluating false positives and checking detected entities for errors. This requires more extensive labeling (to check that a detection is absent from that list) or more detailed logic (to conclude that the detection is not a person). For instance, it is undesirable to penalize NER for identifying an person entity from cues when the surname is not present (e.g. "Votre candidat député").

## 5 Results and discussion

### 5.1 Descriptive statistics

We consider manifestoes for the 1988 "législatives" election for which both the main candidate name and the transcript are available. In total, we have 3460 such transcripts.

As we want to study political affiliation, we further restrict the dataset to transcripts with at least one non-independent party in the `soutien` list. This ensures that the political position computed in Section 2.1 is not null and results in a final set of 1986 transcripts.

In the 1986 transcripts, we identify 5140 appearances of the name of the main candidate. Names are repeated on average 2.6 times. Some names appear many times in a transcript (up to 25 times), but this is quite rare as 90% of the names are repeated less than 4 times. As shown in Table 2, candidates and political parties are not significantly over-represented, so we retain the 5140 appearances for our analysis.

Table 2: Counts of name appearances by affiliation. In 374 left-wing transcripts, the main candidate name appears twice.

| Number of repeats | Left | Right |
|---:|---:|---:|
| 1 | 156 | 121 |
| 2 | 374 | 703 |
| 3 | 160 | 156 |
| 4 | 84 | 75 |
| 5 | 24 | 32 |
| ⋮ | ⋮ | ⋮ |
| 15 | 1 | 2 |
| 16 | 1 | 1 |
| 17 | 0 | 1 |
| 18 | 0 | 2 |
| 25 | 0 | 1 |

### 5.2 PER / LOC / ORG tasks

Overall, GLiNER2 retrieves 1886 out of 5140 names, i.e. 0.37 recall. This is a bit lower but consistent with the performance obtained during the labs on the AdminSet dataset [8] for zero-shot learning.

Regarding misclassifications of names, only 5 appearances are extracted as locations and 2 as organizations, which is a very small number. There is no apparent pattern in these extractions. We consider that GLiNER2 solves this question and do not investigate further.

GLiNER2 seems to distinguish very well between names of persons and other entities. This is surprising because the names do not correspond to very famous people who would appear in GLiNER2's pretraining dataset. We deduce that names are recognized based on linguistic features (syntax, semantics) rather than memorization of well-known persons.

### 5.3 Classification task

In Section 4.1, we ask GLiNER2 to classify the manifestos into "left" and "right" categories. Figure 2 illustrates the distribution of the true position index for both predicted classes. The results show a clear separation, indicating correct classification into left-wing and right-wing for manifestos with extreme values. However, center-right manifestos with position around 0.6 are sometimes classified as left or right, suggesting that classification into three classes may more accurately describe the transcripts.

To provide a numerical assessment of the prediction performance, we exclude the center-right class (index between 0.51 and 0.9)[2], and report the classification metrics in Table 3. GLiNER2 correctly infers the political affiliation for polarized transcripts and matches the performance of Doc2Vec.

### 5.4 Extraction of detailed entities

We now evaluate the performance of the `PER_LEFT`, `PER_RIGHT`, `PER_GAUCHE` and `PER_RIGHT` entities. We first compare their recall with the simple `PER` specification. Table 4 shows that we obtain significantly fewer detections when the task is specified with `PER_LEFT` and `PER_RIGHT` than with `PER` (150+290 is much lower 1886). Specifying the task in French does not improve this.

[2] It may come as surprising that a quarter of the dataset corresponds to the extreme right, i.e. position 1, and the extreme left is absent. This is a consequence of the position definition. Around a quarter of the candidates are supported by the Front National and positioned at 1. The communist party is instead classified as left-wing but not extreme, and is thus positioned at 0.2, which also explains the high number of left-wing name appearances.

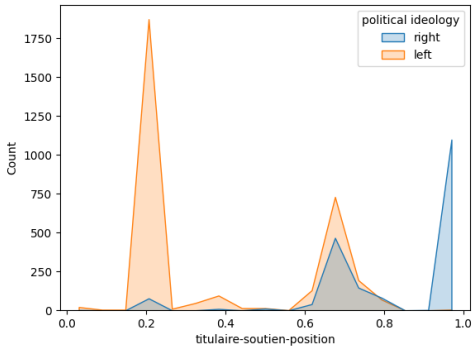


Figure 2: Distribution of the true position index for the predicted political affiliations. GLiNER2 correctly identifies manifestos with position 0 as belonging to the left-wing and position 1 as belonging to the right wing. Center-right manifestos are sometimes classified as left or right.

Table 3: Classification report for the left/right entity labels

| **Label** | **Precision** | **Recall** | **F1-score** | **Support** |
|---|---|---|---|---|
| left | 1.00 | 0.95 | 0.98 | 2180 |
| right | 0.92 | 1.00 | 0.95 | 1103 |
| **accuracy** | | | 0.97 | 3283 |
| **macro avg** | 0.96 | 0.98 | 0.97 | 3283 |
| **weighted avg** | 0.97 | 0.97 | 0.97 | 3283 |

We restrict the task to the correct class by considering only names with position equal to 0.2 and compute the recall for the `PER_LEFT` entity. We obtain 0.063 instead of 0.029, which means that the `PER_LEFT` entity performs better on left-wing transcripts, i.e. the result is correct, but the performance is still very poor compared to the expected 0.37 recall of the simple task. For the `PER_RIGHT` entity, performance does not increase, which suggests that recognition of the right-wing entities is not very good. These results are confirmed by the histograms in Figure 3, which show that the detected `PER_LEFT` entities are concentrated on true left-wing transcripts whereas the detected `PER_RIGHT` entities are not concentrated on right-wing transcripts.

When comparing with the `PER_GAUCHE` and `PER_DROITE` entities, we observe that recall improves for `PER_GAUCHE` but drastically falls for `PER_DROITE`. Therefore, the effect of specifying the task in French is inconsistent. It could also mean that the representations that were learned by GLiNER2 are better described in English than in French although the model can process texts in both languages.

Lastly, we compute the overlap, i.e. the number of candidate names detected by two entities. The `PER_LEFT` and `PER_RIGHT` entities are well separated and only one common occurence is found. `PER_GAUCHE` and `PER_DROITE` overlap (11 common out of 45 `PER_DROITE` entities). We also observe that `PER_LEFT` and `PER_GAUCHE` have only a small overlap (40 common names), which suggests that they retrieve different real-world concepts. However, it is difficult to understand if the mismatches are be related to low detection or to differences between the entity meanings.

In conclusion, using the detailed entities leads to much poorer performance than the standard PERSON extraction. We can attribute the poor performance to the problem specification as a detailed NER task rather than NER or classification capabilities thanks to the baselines established in Section 5.2 for NER and in Section 5.3 for GLiNER2's classification performance. This result also suggests that the cues used in classification are not localized or closely related to specific entity mentions.

### 5.5 Structured extraction

Finally, we explore the performance of the structured extraction task. In this task, we ask GLiNER2 to extract all the persons mentioned in the transcript and provide their name and their political affiliation.

Table 4: Entity-presence statistics

| | PER_LEFT | PER_GAUCHE | PER_RIGHT | PER_DROITE |
|---|---|---|---|---|
| number of entities | 150 | 216 | 290 | 45 |
| recall | 0.029 | 0.042 | 0.056 | 0.009 |
| within correct class | 123 | 168 | 57 | 8 |
| recall | 0.063 | 0.086 | 0.052 | 0.007 |

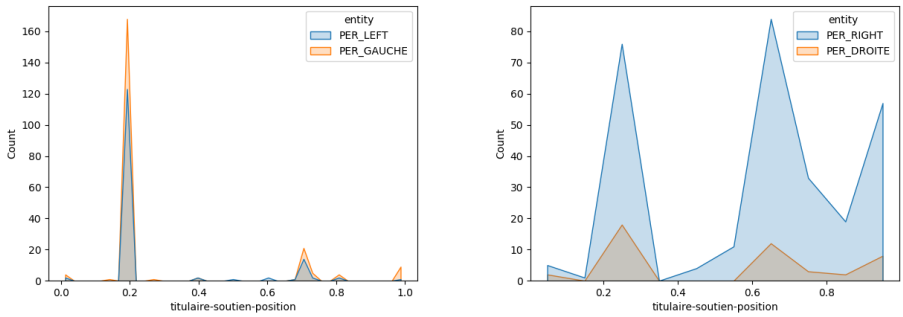


Figure 3: Histograms of the detected entities. `PER_LEFT` and `PER_GAUCHE` entities are detected in left-wing transcripts, whereas `PER_RIGHT` and `PER_DROITE` entities are less detected and less concentrated.

GLiNER2 extracts 4510 persons in total, which includes the main candidate, the name of the running mate as well as other individuals. For each transcript, we consider that GLiNER2 is successful if the name of the main candidate is inside one the names of the persons extracted from the transcript.

The extracted structures occasionally contain null values. We excluded 276 persons with a null name as we cannot recognize the main candidate from these extractions. 12 persons have a name but a null affiliation although the schema only allows `left` and `right`. This happens once for a main candidate and 7 times for the running mate and is sufficiently rare to be excluded from our analysis.

We hypothesized that structured extraction would be more challenging than NER, as it involves analysing the complete document as well as extracting a complex object. Contrary to our expectations, GLiNER2 identifies the name of the main candidate in 1612 out of the 1986 transcripts, i.e. a recall of 0.81[3]. This performance is stable across political affiliation, although right-wing candidate names are extracted less frequently (0.73 instead of 0.84).

This value cannot be directly compared to the 0.37 recall because it is at transcript level instead of entity level. Instead, we compare structured extraction to the performance of recalling at least one occurence of the candidate name. In transcripts with repeats, we expect the task to be easier because the name can be extracted from the second occurrence even if the first occurrence was not detected as an entity. In Figure 4, we plot the recall of the NER task and the structured extraction task as a function of the number of repeats. As expected, both increase with more repeats, and structured extraction performance is lower because the task is harder.

Figure 4 also illustrates the probabilty of recalling each entity as a function of the number of occurence. This highlights a puzzling phenomenon. When the candidate name is present only once, NER recall is 0.81, slightly higher than structured recall (0.73). We expected that when the name is present multiple times, GLiNER2 would maintain similar performance across repeats. This is not the case, and GLiNER2 only detects one of them in most transcripts.

The reason is GLiNER2 only detects entities once, even when asked for the corresponding span, so repeats are usually ignored. When the name is repeated in different forms, GLiNER2 reports multiple entities, likely because they can be interpreted as distinct people: different first names `{"Suzanne SAUVAIGO",66,82}`, `{"Marcia S. SAUVAIGO"}`, accent forms

[3]To compute a recall for the name instead of a span, we check if the normalized name is a substring of the extracted name

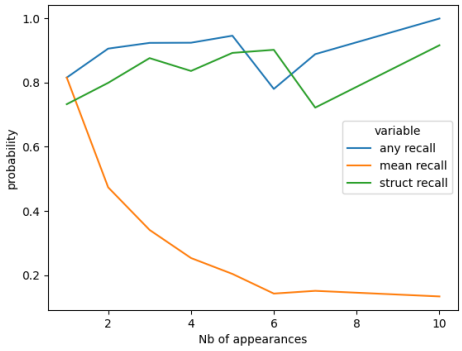


Figure 4: Transcript level performance of NER and structured extraction tasks by number of occurrences. Performance increases with the number of occurrences and is lower for structured extraction. However, the probability of recalling each entity falls because GLiNER2 only detects the first occurrence.

`{"Gérard MOREAU",446,459}`, `{"GERARD MOREAU",30,43}`[4]. We provide a minimal working example in the source code. This behavior is not mentioned in the model documentation but has been reported in a https://github.com/urchade/GLiNER/issues/242. It may also explain why we obtained lower performance compared to spaCy during the course labs.

Finally, we evaluate the classification performance for the political affiliation, for the candidates that were correctly recalled. Table 5 shows the classification report when only considering the `left` and `right` categories as in Table 3. As expected, we obtain a very good, but slighty lower F1-score.

Table 5: Classification report for the "left"/"right" labels.

| **Label** | **Precision** | **Recall** | **F1-score** | **Support** |
|---|---|---|---|---|
| left | 1.00 | 0.91 | 0.95 | 708 |
| right | 0.86 | 0.99 | 0.92 | 393 |
| **accuracy** | | | 0.94 | 1101 |
| **macro avg** | 0.93 | 0.95 | 0.94 | 1101 |
| **weighted avg** | 0.95 | 0.94 | 0.94 | 1101 |

## 6 Conclusion

In this project, we investigate the extraction of candidate names and political affiliations from manifesto transcripts. In particular, we are interested in specifying this as a structured extraction task to facilitate large-scale deployments at INSEE. To this end, we chose to evaluate several possible task specifications in the GLiNER2 framework.

We obtain several insights which will be helpful for future applications. First, GLiNER2 does not detect duplicate entities and may be more suited for document-level analysis than exhaustive entity level analysis. Second, detailed entities combining a person with a political affiliation such as `PER_LEFT` suffer from significantly lower recall, even though we provided a more detailed specification. Third, the language used in the specification can affect the results, but prompting in English remains an effective method to process French documents. Finally, structured extraction provides well-formatted, machine-readable results with only a marginal loss in performance.

The current analysis could be extended in several ways. We did not optimize the clustering of party affiliations because the computational burden was high, even though we selected a lightweight model. Future work could improve the model performance by providing examples (few-shot learning) or training the model for specific context. Recent advances in output format enforcement methods

[4]We trained GLiNER2 on the raw, unnormalized transcripts.

could enhance the reliability of large language models[5]. We believe that these refinements would not contradict our main conclusions.

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[5] e.g. https://docs.ollama.com/capabilities/structured-outputs